

Antonio Carmagnani

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PROFILE

Experience in computational design and data analysis for real estate and cities, spanning academic research and architecture practice. Recently completed an M.Arch. in Architecture Computation at The Royal Danish Academy, with a focus on computational and data methods for real estate and urban development. Hands-on skills span Python, QGIS, and machine learning, parametric workflows in Grasshopper, and building data tools that turn spatial and financial data into decisions. Motivated by work where the built environment, technology, and data intersect.

EXPERIENCE

Project Assistant in International Infrastructure Projects @ Sweco Danmark A/S

Copenhagen, DK

02.2026 - 07.2026

- Support consulting teams delivering internationally funded infrastructure projects in water, energy, and transport across emerging and developing economies, financed by development banks including the World Bank, EBRD, IFDK, KfW and EIB.
- Built geospatial databases for water and infrastructure projects in QGIS, compiling georeferenced asset data, satellite imagery, and open sources for socio-spatial analysis.
- Developed a pricing tool to support bid decisions, parametrizing expert fee bands and project economics for financial viability.
- Contributed to the technical and methodology sections of tender proposals, articulating Sweco's delivery approach for infrastructure project bids.
- Engaged national water agencies and government counterparts during the field missions to align on rehabilitation priorities and collect technical and geographic project data.

Intern in Responsible Business Conduct @ Ramboll Management Consulting

Copenhagen, DK

05.2025 - 08.2025

- Conducted human rights due diligence for corporate clients, applying the UN Guiding Principles on Business and Human Rights and OECD Guidelines to assess salient risks across global supply chains.
- Built a fully automated risk-assessment system in Excel, consolidating risk data across supply chains into a single dashboard that updated salient risks dynamically.
- Integrated AI research tools into due diligence workflows to source and cite human rights risk data at scale across high research volumes.
- Led an internal market study mapping 20+ critical raw materials supply chains to identify prospective clients and new ESG service lines, connecting regulatory drivers to concrete business development targets.
- Ran a peer benchmarking analysis for a corporate client, comparing its responsible business and human rights practices against industry competitors across multiple service areas to identify gaps, competitive advantages, and opportunities to strengthen its positioning.

Architect @ Isay Weinfeld Architects

São Paulo, BR

02.2023 - 08.2024

- Delivered high-end residential, hotel, and mixed-use projects across South America, North America, Europe, the Middle East, and Asia at one of Brazil's most prominent architecture studios.
- Built a Grasshopper plugin in Python that generates bespoke staircases from a floor plan, cutting staircase modeling from days to around 20 minutes and enabling far more design iterations and geometric complexity.
- Presented the tool in a studio workshop and supported its adoption by colleagues across 10+ projects.
- Scripted parametric and computational models in Rhino and Grasshopper for geometrically complex forms, supporting design development across projects.

Trainee in Real Estate Product Development @ OSPA

São Paulo, BR

02.2022 - 10.2022

- Produced 15+ feasibility studies for mixed-use and residential developments across Brazil and the US (São Paulo, Rio de Janeiro, Porto Alegre, Miami) for developer clients.
- Ran financial feasibility studies on candidate sites, sizing investment and return against buildable area, unit mix, price per square metre, and parking to test viability.
- Produced images, plans, and diagrams and presented feasibility outcomes to developer clients.

Architect @ Felipe Hess Architects

São Paulo, BR

07.2021 - 02.2022

- Developed high-end residential projects in São Paulo and Rio de Janeiro from preliminary studies through technical documentation, working in Revit, AutoCAD, and SketchUp.

Intern in Project Management @ Itaú-Unibanco

São Paulo, BR

08.2020 - 07.2021

- Reviewed technical documentation from architecture and engineering firms prior to construction authorization, checking drawings against building, accessibility, and fire-safety norms (NBR standards).
- Built a VBA-powered Excel dashboard tracking the bank's corporate furniture inventory across locations, replacing a manual process and speeding up procurement for new-building fit-outs.

Independent Consultant @ Self-employed

São Paulo, BR and Copenhagen, DK

07.2018 - Present

- Delivered independent consulting to clients across the AEC sector, spanning product and UX advice, Revit and Dynamo automation, and financial analysis tooling.
- Co-founded and ran a small practice delivering projects for 10+ clients in the AEC sector, responsible for client acquisition, scoping, pricing, and delivery.
- Advised a software startup in the built environment over two years on product direction, UI and UX, and applied AI, testing releases and returning improved Revit and Dynamo scripts.
- Built an Excel and VBA financial analysis tool for a private client that ingests monthly statements and PDF exports, categorises transactions automatically, and renders them into a decision dashboard.

EDUCATION	M.Arch. in Architecture Computation @ The Royal Danish Academy	Copenhagen, DK
	07.2024 - 07.2026	
	Semester Programme in Building Technology @ Architecture Association	London, UK
	07.2022 - 01.2023	
	Summer School in Parametric Design @ Architecture Association	London, UK
	06.2019 - 07.2019	
	B.Arch. in Architecture and Urbanism @ Mackenzie University	São Paulo, BR
	07.2018 - 12.2023	
PROJECTS	Master's Thesis, Computational Urban Analysis for Intensification Strategy @ The Royal Danish Academy	Copenhagen, DK
	- Developed a computational framework to identify underused spaces in the historic city center of São Paulo, Brazil, combining urban morphology analysis, land use data, and social vulnerability indicators. - Built an algorithm that uses urban data, including building typology, use patterns, regulatory conditions, and building use, to generate programmatic and morphological typologies for urban intensification, aligned with the city's existing urban policies. - Integrated multiple geospatial datasets in QGIS to map spatial relationships between buildings, zoning conditions, and social context at the neighborhood scale.	
	House Price Prediction - Competition	Copenhagen, DK
	Built a predictive model for real estate valuation using ensemble learning (XGBoost, LightGBM, Ridge, CatBoost) and feature engineering on structured property and financial data, finishing in the top 2% of competitors.	
	Real Estate Development Recommendation Tool @ The Royal Danish Academy	Copenhagen, DK
	- Built a machine learning tool that generates site-specific development recommendations for buildings in Copenhagen by integrating financial, social, sustainability, and geospatial data. - Designed a user interface allowing different stakeholders to adjust impact drivers and weighting, with predictive outputs aligned to ESG priorities and local development conditions.	
SKILLS	Computational Deployable Structures Workshop @ Southeast University x The Royal Danish Academy	Nanjing, CN
	- Selected as one of a small group from The Royal Danish Academy for a two-week international workshop at Southeast University, applying origami-based deployable-structure principles (used in satellite deployment) to architectural structures. - Developed parametric models exploring foldable geometry, material systems, and structural performance, working across computational design, geometry, and engineering.	
	- Data & Analysis: Python, QGIS, SQL, Power BI, Excel / VBA, HTML - Digital Tools: Rhino + Grasshopper, Revit, AutoCAD, Adobe Suite (InDesign, Illustrator, Photoshop) - Real Estate & Urban: Real estate feasibility modeling, Zoning and land use regulatory analysis, Urban morphology and spatial analysis - Languages: English (fluent), Portuguese (native), Spanish (fluent), Danish (limited working proficiency), Italian (limited working proficiency)	